COMPETENCY STANDARD

FOR

INDUSTRIAL ENGINEERING AND LEAN MANUFACTURING

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Industrial Engineering and Lean Manufacturing (RMG sector) is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP).** This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-IELM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1 Identify OHS policies and procedures. 2 Apply personal health and safety practices. 3 Report hazards and risks. 4 Respond to emergencies	08
SICIP-RMG-IELM-02-G	Manage personal and professional development	1 Interpret personal development skills. 2 Set and meet self-development priorities. 3 Maintain professional growth and development.	10
SICIP-RMG-IELM-03-G	Support innovation and manage change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	12
		Total Hour	**30**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-IELM-01-S	Apply Quality Procedures	1. Identify quality procedures 2. Follow quality procedure 3. Maintain standard procedure	10
SICIP-RMG-IELM-02-S	Apply green practices in RMG sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**20**

Occupation Specific Competencies (222 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-IELM-01-O	Identify basic garment construction	1. Interpret manufacturing process. 2. Identify industrial sewing machine and functions. 3. Identify stitches and seams. 4. Record clothing materials	20
SICIP-RMG-IELM-02-O	Perform work study	1. Define work study. 2. Define method study. 3. Carry out work measurement. 4. Perform SMV calculation. 5. Perform production capacity and target calculation. 6. Perform efficiency calculation	72
SICIP-RMG-IELM-03-O	Perform line layout	1. Carry out garment's operation breakdown. 2. Identify types of layouts 3. Perform line layout.	28
SICIP-RMG-IELM-04-O	Implement lean manufacturing system	1. Define lean manufacturing 2. Identify lean manufacturing waste 3. Apply lean tools. 4. Interpret six sigma	52
SICIP-RMG-IELM-05-O	Apply tools for ensuring quality	1. Identify tools of quality 2. Identify garment defects 3. Use tools for rectifying the defects	12
SICIP-RMG-IELM-06-O	Perform production planning and optimization	1. Interpret TNA plan. 2. Perform plant capacity calculation. 3. Prepare inventory plan. 4. Prepare production schedule 5. Perform clothing material utilization. 6. Perform process optimization	38
		Total Hours	**222**

COMPETENCY STANDARD: INDUSTRIAL ENGINEERING AND LEAN MANUFACTURING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 08 hrs.	Unit Code: SICIP-RMG-IELM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including **<u>personal protective equipment (PPE).</u>** 2.2 Common health issues are recognized. 2.3 Common safety issues are identified.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 **<u>Emergency response plans and procedures</u>** are responded to. 4.3 **<u>First aid procedures</u>** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Emergency response plans and procedures	1.1 Firefighting procedures 1.2 Earthquake response procedures 1.3 Emergency response plans and procedures 1.4 Medical and first aid
3. First aid procedure	1.1 Washing of open wound 1.2 Washing chemically infected area 1.3 Applying bandage 1.4 Taking appropriate medicine
4. Personal protective equipment	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors

	in the workplace
4 Resource implications	The following resources must be provided: 3.1 Workplace (simulated or actual) 3.2 Personal protective equipment (PPE) 3.3 Firefighting equipment 3.4 Emergency response manual 3.5 First aid kits 3.6 Stationery 3.7 Learning manual.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MANAGE PERSONAL AND PROFESSIONAL DEVELOPMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-IELM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to manage personal and professional development. It specifically includes the tasks of interpreting personal development skills, setting and meeting self-development priorities and maintaining professional growth and development.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret personal development skills	1.1 Objectives of personal development skills are described. 1.2 **<u>Personal development skills</u>** are identified. 1.3 Intra and Interpersonal relationships are maintained in the course of managing oneself. 1.4 Self-analysis is performed and personal development needs are identified.
2. Set and meet self-development priorities	1.5 Tasks are prioritized to achieve personal, team and organizational goals and objectives. 1.6 **<u>Resources</u>** are utilized efficiently and effectively to manage work priorities and commitments. 1.7 Economic usage and maintenance of facilities are followed as per established procedures.
3. Maintain professional growth and development	1.8 Pro activeness/zeal is demonstrated in fulfilling personal and professional growth requirements. 1.9 **<u>Trainings and career opportunities</u>** are identified and accessed based on job requirements. 1.10 **<u>Recognitions</u>** are sought/ received and demonstrated as proof of career advancement. 1.11 **<u>Licenses and/or certifications</u>** relevant to the job and career are obtained and renewed

Range of variables:

Variables	**Range** (may include but not limited to)
1. Personal Development skills	1.1 Problem-solving 1.2 Self-confidence 1.3 Adaptability 1.4 Integrity 1.5 Work ethic 1.6 Pro-activeness

2. Resources	1.7 Human 1.8 Financial 1.9 Technology
3. Trainings and career opportunities	1.10 Participation in training programs 3.1.1 Technical 3.1.2 Supervisory 3.1.3 Managerial 3.1.4 Continuing Education 1.11 Serving as Resource Persons in conferences and workshops
4. Recognitions	1.12 Recommendations 1.13 Citations 1.14 Certificate of Appreciations 1.15 Commendations 1.16 Awards 1.17 Tangible and Intangible Rewards
5. Licenses and/or certifications	1.18 National Certificates 1.19 Certificate of Competency 1.20 Support Level Licenses 1.21 Professional Licenses

Curricular Content Guide

1. Underpinning knowledge	1.1 Importance of personal development skills. 1.2 Organizational policies relevant to training and professional growth. 1.3 Company operations, procedures and standards. 1.4 Resources in work environment.
2. Underpinning skill	1.5 Utilizing and improving personal development skills 1.6 Maintaining Intra and Interpersonal relationship 1.7 Utilizing communication skills 1.8 Prioritizing tasks in accordance with work commitment. 1.9 Utilizing resources efficiently and effectively
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.1 Workplace (simulated or actual) 1.2 Workplace procedures 1.3 Standard operating procedure

	1.4 Workplace documents, signs and symbols 1.5 Codes of conduct 1.6 Projector 1.7 Stationery 1.8 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 attained personal development skills. 1.2 maintained intra and interpersonal relationship in the course of managing oneself. 1.3 prioritized tasks according to work commitments. 1.4 identified training and career opportunities. 1.5 completed trainings based on the requirements of the industries. 1.6 acquired and maintained licenses and/or certifications according to the requirement of the qualification.
2. Methods of assessment	Competency should be assessed by: 1.7 Written test 1.8 Practical Demonstration 1.9 Oral Questioning 1.10 Portfolio (Optional)
3. Context of assessment	1.11 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.12 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SUPPORT INNOVATION AND MANAGE CHANGE**	**Nominal Duration:** 12 hrs.	**Unit Code:** SICIP-RMG-IELM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approach and solution, supporting flexible and innovative ways of working and adapting to emerging changes and opportunities.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for **<u>innovation</u>** in scope of work is recognized 1.2. The value of **<u>innovative practices</u>** in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1 Opportunities and creative approaches to implement innovation practices are identified 2.2 Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3 Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1 Maximum innovative opportunities are promoted. 3.2 Work assignments to facilitate innovative work skills are organized. 3.3 Team members are provided with guidance and coaching on innovation in the workplace. 3.4 Models of innovative work practice are provided and discussed 3.5 Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1 Usages of different **<u>technologies</u>** is determined based on job requirements. 4.2 Appropriate technology is selected as per work specifications. 4.3 Relevant technology is effectively used in carrying out functions. 4.4 Appropriate implementation tools are used as per task

	requirement.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	1.4. New ideas 1.5. Different ideas 1.6. New methods of doing work 1.7. Use of new Technology
3. Technologies	1.8. Office technology 1.9. Industrial technology 1.10. System technology 1.11. Information technology 1.12. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization’s policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization’s operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	1.1. Identifying resources required for creativity and innovation 1.2. Contributing to brainstorming session 1.3. Identifying issues and concerns of one’s scope of work 1.4. Encouraging co-workers to generate and develop ideas 1.5. Evaluating potential obstacles to and opportunities for

	creativity and innovation 1.6. Sharing of best practices related to innovation and creativity 1.7. Applying relevant technology 1.8. Using update machine and Software 1.9. Acquiring troubleshooting skills Identifying issues 1.10. Identifying current industry standard diagnostic tools 1.11. Describing common malfunctions and resolutions. 1.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the workplace 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **APPLY QUALITY PROCEDURES**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-IELM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply quality procedures. It specifically includes the tasks of identifying quality procedures, following quality procedure and maintaining standard procedure.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify quality procedures	1.1. **<u>Manuals</u>** and guidelines for RMG production processes are collected as per sample specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedure	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with demand of **<u>quality improvement system.</u>** 2.2 Concept of supplying product or service to meet the **<u>customer quality requirements</u>** is understood and accordingly applied. 2.3 Conformance to specifications is ensured. 2.4 Defects are detected and reported to the relevant authority according to standard operating procedures
3. Maintain standard procedure	3.1 Performance is assessed at regular interval. 3.2 Specifications and standard operating procedures are established 3.3 Quality of product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in every case at all situations.

Range of Variables

Variable	Range May include but not limited to:
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual

	1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. quality improvement system.	2.1. Quality inspection 2.2. Testing 2.3. Quality control. 2.4. Quality assurance 2.5. Total Quality Management
3. customer quality requirements	3.1. Performance 3.2. Features 3.3. Reliability 3.4. Conformance 3.5. Aesthetics 3.6. Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Importance of maintaining quality 1.2 quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 1.3 Process and procedures for improving and maintaining quality 1.4 Procedures for addressing defects. 1.5 Record keeping within the quality improvement system in workplace 1.6 Factors, which affect successful implementation of the quality systems and procedures.
2. Underpinning Skills	2.1. Maintaining good quality 2.2. Eliminating poor quality 2.3. Understanding the meaning of the key terms - quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 2.4. Improving and maintaining quality 2.5. Addressing defects and procedures 2.6. Recording within the quality improvement system in workplace. 2.7. Implementing quality systems and procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified quality procedures 1.2 followed quality procedure 1.3 maintained standard procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN RMG SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-IELM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices in RMG (Ready-Made Garment)</u>** manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 **<u>Sources of environmental impacts</u>** during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 **<u>Ways of mitigating environmental impacts</u>** in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of waste in RMG manufacturing</u>** are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation.

	1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 **Recyclable Items in RMG Manufacturing** 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in RMG Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products.

	5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices

2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY BASIC GARMENT CONSTRUCTION	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-IELM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify basic garment construction. It specifically includes the tasks of interpreting manufacturing process, identifying industrial sewing machines and functions, identifying stitches and seams, and recording clothing materials.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret manufacturing process	1.1 **Fabric manufacturing process** is identified and explained. 1.2 **Garment manufacturing process** is identified and explained.
2. Identify industrial sewing machines and functions	2.1 **Types of industrial sewing machine** are identified. 2.2 Functions of industrial sewing machine are identified and described. 2.3 Types of **sewing needles** are identified. 2.4 Types of **attachments** used with industrial sewing machine are identified.
3. Identify stitches and seams	3.1 Types of **stitches** are identified as per garment style. 3.2 Types of **seams** are identified as per garment style. 3.3 Garment stitch quality is interpreted as per sample. 3.4 Garment seam quality is interpreted as per sample.
4. Record clothing materials	4.1 Different types of **clothing materials** are identified. 4.2 **Bill of materials (BOM)** is identified. 4.3 Clothing materials are itemized and recorded as per the BOM (Bill of Material) sheet.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Fabric manufacturing process	1.1 Fiber 1.2 Yarn 1.3 Woven fabric 1.4 Knit fabric 1.5 Dying 1.6 Printing 1.7 Finishing 1.8 Yarn dyed fabrics
2. Garment manufacturing	2.1 Design

process	2.2 Pattern making 2.3 Fit sample making 2.4 Production pattern making 2.5 Grading 2.6 Marker making 2.7 Fabric spreading 2.8 Fabric cutting 2.9 Cutting parts numbering and bundling 2.10 Sewing 2.11 Washing and embellishment 2.12 Finishing and packaging
3. Types of Industrial sewing machine	3.1 Single needle (chain and lock) 3.2 Double needle (chain and lock) 3.3 Over lock 3.4 Flat Lock 3.5 Feed of the arm 3.6 Multi needle chain stitch 3.7 Two step zigzags 3.8 Three step zigzags 3.9 Bar tack 3.10 Button hole 3.11 Button stitch 3.12 Eyelet
4. Sewing needles	4.1 DBx1 (Versatile for general stitching on medium-weight woven fabrics) 4.2 DBx5 (Designed for high-speed, heavy-duty straight stitching) 4.3 DCx1 (Used for parallel stitching on various fabric types) 4.4 DCx5 (Heavy-duty double-needle stitching for thick materials) 4.5 DCxK5 (For stretchy fabrics, like knits and lycra, using double-needle setup)
5. Attachments	5.1 Feed 5.2 Guide 5.3 Folder
6. Stitches	6.1 Chain 6.2 Lock 6.3 Hand
7. Seams	7.1 Super imposed 7.2 French 7.3 Lapped 7.4 Lap felled 7.5 Bound 7.6 Flat 7.7 Decorative 7.8 Edge neatening
8. Clothing materials	8.1 Fabric

	8.2 Trims and accessories
9. Bill of materials (BOM)	9.1 Item Description 9.2 Material Specifications 9.3 Quantity Required 9.4 Unit of Measurement 9.5 Garment Parts 9.6 Trim and Accessories 9.7 Garment Size Information

Curricular Content Guide

1. Underpinning Knowledge	1.1 Steps from fiber to finished fabric: spinning, weaving/knitting, dyeing, finishing. (Fabric manufacturing process) 1.2 Processes like designing, pattern making, cutting, sewing, finishing, and packing. (Garment manufacturing process) 1.3 Machines like lockstitch, overlock, flatlock with attachments for specific operations. (Industrial sewing machines (including attachments) 1.4 Needles vary by size, point type, and machine compatibility for different fabrics (Sewing needles) 1.5 Different types such as lockstitch, chainstitch, overlock; seems like plain, lapped, bound (Stitches and seams) 1.6 Materials include fabrics, linings, interlinings, trims, accessories (clothing materials) 1.7 A detailed list of all materials and components needed to produce a garment (Bill of materials)
2. Underpinning Skills	2.1 Explaining fabric manufacturing process 2.2 Explaining garment manufacturing process 2.3 Identifying industrial sewing machines and their functions 2.4 Identifying types of stitches and seams 2.5 Identifying clothing materials 2.6 Recording clothing materials as per bill of material (BOM)
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Fabric (samples) 4.3 Garment (samples) 4.4 Trims and accessories 4.5 Bill of Materials (samples)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted fabric manufacturing process 1.2 identified industrial sewing machines and functions 1.3 identified stitches and seams 1.4 recorded clothing materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM WORK STUDY**	**Nominal Duration:** 72 Hrs.	**Unit Code:** SICIP-RMG-IELM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform work study. It specifically includes the tasks of defining work study, defining method study, carrying out work measurement, performing SMV calculation, performing production capacity and target calculation and performing efficiency calculation.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define work study	1.1 Work study is defined. 1.2 Importance of work study is explained. 1.3 Basic procedure of work study is identified and described.
2. Define method study	2.1 Method study is defined. 2.2 **<u>Method study procedures</u>** are identified and explained. 2.3 Primary and secondary questioning techniques are identified.
3. Carry out work measurement	3.1 Purpose of work measurement is explained. 3.2 **<u>Work measurement techniques</u>** are identified. 3.3 Work measurement procedure is identified and implemented to assess effectiveness. 3.4 Employee ratings and allowances for work content are identified.
4. Perform SMV calculation	4.1 **<u>Tools for SMV calculation</u>** are identified. 4.2 Procedure for SMV calculation is identified. 4.3 Cycle check data is identified and computed 4.4 **<u>SMV calculation formula</u>** is identified and interpreted. 4.5 SMV calculation is performed according to formula.
5. Perform production capacity and target calculation	5.1 **<u>Production capacity</u>** on process, line and factory are identified. 5.2 Production capacity calculation formula is identified interpreted. 5.3 Production capacity is calculated as per formula. 5.4 **<u>Production target calculation formula</u>** is identified and interpreted. 5.5 Production target is calculated as per formula.
6. Perform efficiency calculation	6.1 **<u>Efficiency calculation method</u>** is identified. 6.2 Efficiency calculation formula is interpreted. 6.3 Efficiency calculations are prepared according to formula.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Method study procedures	1.1 Select 1.2 Record 1.3 Examine 1.4 Develop 1.5 Evaluate 1.6 Define 1.7 Install 1.8 Maintain
2. Work measurement techniques	2.1 Work sampling 2.2 Time study 2.3 Predetermined time standards (PTS) 2.4 Standard rate
3. Tools for SMV calculation	3.1 Stop watch 3.2 Paper, pen, pencils 3.3 Calculator 3.4 Computer
4. Standard Minute Value (SMV) calculation formula	4.1 Cycle time 4.2 Observed time 4.3 Basic time 4.4 Performance rating 4.5 Allowances
5. Production capacity	5.1 Working hour 5.2 SMV 5.3 Total SMV earners 5.4 Working days
6. Production target calculation formula	6.1 Production capacity 6.2 Efficiency 6.3 Absenteeism
7. Efficiency calculation method	7.1 Production output 7.2 SMV 7.3 SMV earners 7.4 Working hours

Curricular Content Guide

1. Underpinning Knowledge	1.1 Systematic analysis to improve work efficiency (work study) 1.2 Best way to perform a task safely and efficiently (work method) 1.3 Methods to determine standard time (work measurement techniques) 1.4 Work measurement procedure 1.5 SMV calculation formula 1.6 Production capacity calculation formula

	1.7 Production target calculation formula 1.8 Efficiency calculation formula
2. Underpinning Skills	2.1 Defining work study and work method 2.2 Carrying out work measurement 2.3 Performing SMV calculation 2.4 Performing production capacity calculation 2.5 Performing production target calculation 2.6 Performing efficiency calculation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Calculator 4.4 Projector 4.5 Stationery 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined work study 1.2 defined method study 1.3 carried out work measurement 1.4 performed SMV calculation 1.5 performed production capacity and target calculation 1.6 performed efficiency calculation
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM LINE LAYOUT**	**Nominal Duration:** 28 Hrs.	**Unit Code:** SICIP-RMG-IELM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform line layout. It specifically includes the tasks of carrying out garment's operation breakdown, identifying types of layouts and performing line layout.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out garment's operation breakdown	1.1 **Garments operation breakdown** is identified and explained as per styling. 1.2 Garments operation breakdown is prepared as per production sample.
2. Identify types of layouts	2.1 Importance of layout in production line is explained. 2.2 **Types of layouts** are identified as per job requirements.
3. Perform line layout	3.1 **Line layout** is interpreted as per styling. 3.2 Line balancing is interpreted and maintained. 3.3 Line layout in accordance with the production sample is performed.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Garments operation breakdown	1.1 Style of the garments 1.2 Front part 1.3 Back part 1.4 Assembling part 1.5 Make section
2. Types of layouts	2.1 Straight 2.2 Side by side 2.3 U- shaped 2.4 Modular 2.5 Progressive Bundle System (PBS) 2.6 Unit Production System (UPS)
3. Line layout	3.1 Operation breakdown 3.2 Machine selection 3.3 Operator selection 3.4 Standard Minute Value (SMV)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Detailed listing of all steps required to make a garment (garments operation breakdown) 1.2 Arrangement of machines and workers for smooth production flow (line layout) 1.3 Distributing work evenly across operators to reduce idle time and improve output. (line balancing) 1.4 Types include straight-line, U-shape, modular, and side-by-side layouts (layout types)
2. Underpinning Skills	2.1 Performing garments operation breakdown 2.2 Identifying and maintaining line balancing 2.3 performing line layout
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure (samples) 4.3 Projector 4.4 Stationery 4.5 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 performed garments operation breakdown 1.2 identified types of layouts 1.3 performed line layout.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IMPLEMENT LEAN MANUFACTURING SYSTEM**	**Nominal Duration:** 52 Hrs.	**Unit Code:** SICIP-RMG-IELM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to implement lean manufacturing system. It specifically includes the tasks of defining lean manufacturing, identifying lean manufacturing waste, applying lean tools and interpreting six sigma.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define lean manufacturing	1.1 **<u>Basic lean manufacturing systems</u>** are identified. 1.2 Lean manufacturing system is interpreted to increase efficiency.
2. Identify lean manufacturing waste	2.1 **<u>Types of waste</u>** in lean manufacturing are identified. 2.2 Lean manufacturing wastage solutions are identified.
3. Apply lean tools	3.1 Basic lean manufacturing **<u>tools and techniques</u>** are identified. 3.2 Impact of application of tools and techniques to lean manufacturing is described. 3.3 Lean manufacturing tools are applied to enhance organizational productivity and efficiency.
4. Interpret six sigma	4.1 Six sigma is identified and interpreted. 4.2 **<u>DMAIC Framework</u>** is identified and interpreted.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Basic lean manufacturing systems	1.1 Value 1.2 Value stream mapping 1.3 Flow 1.4 Pull 1.5 Perfection
2. Types of waste	2.1 Over production 2.2 Over processing 2.3 Excess transportation 2.4 Excess inventory 2.5 Excess motion 2.6 Waiting 2.7 Re- work 2.8 Unused talents 2.9 Dis-connectivity
3. Tools and techniques	3.1 Value steam mapping (VSM)

	3.2 Workplace organization 3.3 Visual management 3.4 Kanban and super market 3.5 Standardization of work process 3.6 Root cause analysis 3.7 Cellular manufacturing 3.8 SMED 3.9 5S 3.10 Problem solving 3.11 TPM 3.12 Kaizen
4. DMAIC Framework	4.1 Define 4.2 Measure 4.3 Analyze 4.4 Improve 4.5 Control

Curricular Content Guide

1. Underpinning Knowledge	1.1 Lean manufacturing systems 1.2 Seven wastes: overproduction, waiting, transport, extra processing, inventory, motion, defects. 1.3 5S, Kaizen, Value Stream Mapping (VSM), Kanban, Just-In-Time (JIT). 1.4 Methodology to improve quality by reducing defects and variability (six sigma) 1.5 Structured Six Sigma process: Define, Measure, Analyze, Improve, Control (DMAIC) framework
2. Underpinning Skills	2.1 Implementing lean manufacturing system 2.2 Identifying types of lean manufacturing waste 2.3 Applying tools and techniques to organizational production 2.4 Interpreting six sigma
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Projector 4.4 Stationery 4.5 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined lean manufacturing 1.2 identified lean manufacturing waste 1.3 applied lean tools 1.4 interpreted six sigma
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY TOOLS FOR ENSURING QUALITY**	**Nominal Duration:** 12 Hrs**.**	**Unit Code:** SICIP-RMG-IELM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply tools for ensuring quality. It specifically includes the tasks of identifying tools of quality, identifying garment defects and using tools for rectifying the defects.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify tools of quality	1.1 **Basic tools of quality** are identified according to the specific requirements. 1.2 Tools for quality control are selected based on the quality standards to be achieved. 1.3 The usage of tools is aligned with organizational policies.
2. Identify garment defects	2.1 **Garment defects** are inspected and identified according to quality standards. 2.2 Defects are recorded accurately, including the type and location of the defect. 2.3 Defective garments are segregated from non-defective garments to prevent further use.
3. Use tools for rectifying the defects	3.1 Defective garments are examined to determine the appropriate tool required for rectification. 3.2 The defect rectification process is conducted in a manner that minimizes disruption. 3.3 Rectified garments are inspected. 3.4 All rectification actions and tool usage are documented for future reference and quality control.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Basic tools of quality	1.1 Check sheet 1.2 Control chart 1.3 Histogram 1.4 Ishikawa diagram 1.5 Pareto chart 1.6 Scatter diagram 1.7 Flow chart
2. Garment defects	2.1 Fabric defects: 2.1.1 Drop stitches 2.1.2 Dye marks 2.1.3 Laddering 2.1.4 Stains

	2.1.5 Bad selvedge 2.1.6 Shade variation 2.2 Workmanship defects: 2.2.1 Seam puckering 2.2.2 Broken stitches 2.2.3 Open/broken seams 2.2.4 Drop/skipped/stitch 2.2.5 Shading variation 2.2.6 Untrimmed thread 2.3 Trim defects: 2.3.1 Trim broken 2.3.2 Trim differs 2.3.3 Trim bleeding

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic tools like Check Sheet, Histogram, Pareto Chart, Cause-and-Effect Diagram, Control Chart (tools of quality) 1.2 Inspection, testing, auditing, and monitoring processes to maintain standards (quality activities) 1.3 Common defects include stains, broken stitches, uneven seams, wrong measurements. (garment defects) 1.4 Systematic process to ensure products meet required quality standards before delivery (quality assurance)
2. Underpinning Skills	2.1 Identifying basic tools of quality 2.2 Interpreting basic tools of quality as per work order 2.3 Identifying basic quality activities 2.4 Identifying garment defects 2.5 Interpreting garment defects as per sample
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Basic tools of quality (sheets, diagrams, charts) 4.3 Garments (defective samples) 4.4 Work sheets

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified tools of quality 1.2 identified garment defects 1.3 used tools for rectifying the defects
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM PRODUCTION PLANNING AND OPTIMIZATION**	Nominal Duration: 38 Hrs.	Unit Code: SICIP-RMG-IELM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform production planning and optimization. It specifically includes the tasks of interpreting TNA plan, performing plant capacity calculation, preparing inventory plan, performing production schedule, performing clothing material utilization and performing process optimization.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret TNA plan	1.1 **Time and action plan** (TNA plan) is interpreted on basis of lead time. 1.2 TNA plan on a selected order is prepared. 1.3 Planning on critical issues is carried out for inclusion in time schedule.
2. Perform plant capacity calculation	2.1 **Plant capacity calculation** formula is identified. 2.2 Plant capacity formula is interpreted. 2.3 Plant capacity calculation is carried out.
3. Prepare inventory plan	3.1 Importance of inventory planning is explained. 3.2 **Types of inventories** are identified. 3.3 **Material requirement planning** is described. 3.4 Inventory plan is prepared as per production schedule.
4. Prepare production schedule	4.1 **Production scheduling** is explained. 4.2 Production schedule is carried out.
5. Perform clothing material utilization	5.1 Efficiency of **clothing material consumption** is identified. 5.2 Material utilization percentage is calculated as per BOM sheet
6. Perform process optimization	6.1 **Bottle neck process** is identified. 6.2 **Line balancing techniques** are followed. 6.3 **Line balancing tools** are used as per line layout. 6.4 **Balancing loss formula** is interpreted. 6.5 Balancing loss of the line is calculated as per formula.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Time and action plan	1.1 Lead time 1.2 Combined execution 1.3 Cutting 1.4 Sewing

	1.5 Finishing and packing 1.6 Shipment
2. Plant capacity calculation	2.1 Total number of machines 2.2 Total hours factory runs a day 2.3 Total number of workers
3. Types of inventories	3.1 Fabrics 3.2 Trims and accessories 3.3 Finished goods 3.4 Work in process 3.5 Machinery 3.6 Tools and equipment
4. Material requirement planning	4.1 Technical package 4.2 Styling 4.3 Materials type 4.4 Required materials per unit 4.5 Sourcing lead time
5. Production scheduling	5.1 Lead time 5.2 Working days 5.3 Holidays 5.4 Calendar days 5.5 Risk factors
6. Clothing material consumption	6.1 Types of fabric 6.2 Styling of apparel 6.3 Fabric width and length 6.4 Marker efficiency 6.5 Shrinkage of fabric 6.6 Size ratio breaks up 6.7 Trims type 6.8 Accessories type
7. Bottle neck process	7.1 Process SMV 7.2 Capacity 7.3 Capacity utilization 7.4 Idle time 7.5 Work in process 7.6 Set-up time 7.7 Direct labor content 7.8 Direct labor utilization 7.9 Hourly production 7.10 Material supply
8. Line balancing techniques	8.1 Split task 8.2 Share task 8.3 Use parallel work station 8.4 Improving material supply 8.5 Motivation
9. Line balancing tools	9.1 Production sheets 9.2 Daily production report 9.3 Inventory levels by operation

	9.4 Skill matrix 9.5 Capacity study report 9.6 Calculator
10. Balancing loss formula	10.1 Number of allocated machines 10.2 Number of calculated machines

Curricular Content Guide

1. Underpinning Knowledge	1.1 Time and Action plan to track production and delivery deadlines (TNA plans) 1.2 Focus on key risks like material delays, machine breakdowns, quality failures (planning on critical issues) 1.3 Plant Capacity = (Total Operators × Working Hours × Efficiency) / SMV (plant capacity calculation formula) 1.4 Raw materials, work-in-progress (WIP), finished goods (Types of inventories) 1.5 System to plan material needs for timely production (Material requirement planning) 1.6 Planning when and where production activities happen (production scheduling) 1.7 Estimating fabric and trims needed per garment (clothing material consumption) 1.8 Slowest operation that limits overall production flow (bottle neck process) 1.9 Distributing tasks evenly across operators to maximize output (line balancing techniques) 1.10 Line balancing tools 1.11 Balancing Loss (%) = (Sum of Ideal Time / Total Available Time) × 100 (balancing loss formula)
2. Underpinning Skills	2.1 Interpreting TNA plan 2.2 Performing plant capacity calculation formula 2.3 Preparing for inventory planning 2.4 Preparing production schedule 2.5 Calculating material utilization percentage 2.6 Perform process optimization 2.7 Calculate balancing loss of the line
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook

	4.3 TNA Plan (samples) 4.4 Calculator 4.5 Work sheet

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted TNA plan 1.2 performed plant capacity calculation 1.3 prepared inventory plan 1.4 prepared production schedule 1.5 performed clothing material utilization 1.6 performed process optimization
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Industrial Engineering and Lean Manufacturing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
The following machines are required to set up a sewing line as per industry standard		
1.	Single Needle Machine	2
2.	Over Lock Machine	2
3.	Flat Lock Machine	2
4.	Double Needle Machine	1
5.	Feed of the Arm Machine	1
6.	Multi Needle Chain Stitch Machine	1
7.	Bar Tack Machine	1
8.	Button Stitch	1
9.	Button Hole Machine	1
10.	Eyelet Machine	1
11.	Digital Video Camera	1
12.	GSM Cutter	1
13.	GSM Scale	1
14.	Measuring Tape	25
15.	Multimedia Projector with Screen	1
16.	Stop Watch	25
17.	Calculator	10
18.	Desktop Computer/Laptop	5
19.	Printer	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Industrial Engineering and Lean Manufacturing the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.